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Roll No

CS-7004(1)-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Embedded Systems

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1.
 - a) What is an Embedded system? List and define the three main characteristics of Embedded systems.
 - b) What are the main challenges of Embedded computing system design?
2. Discuss the functions and applications of following single purpose processors.
 - a) UART
 - b) Watch Dog Timer
 - c) Keypad controller
3.
 - a) What is Build circuit emulator? Where is it used?
 - b) Write the functions of Linkers and Locators.
4.
 - a) What is Round-Robin software architecture? What are its advantages?
 - b) Explain Embedded software development tools in details.

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5.
 - a) What is Real Time Operating System? How is it different from general purpose OS?
 - b) Compare two scheduling strategies for the real time scheduling - preemptive mode and round robin scheduling.
6.
 - a) With the help of a suitable example explain the concept of Semaphores.
 - b) How to evaluate operating system performance? Give details.
7.
 - a) With the help of a suitable diagram explain the functions of I²C and CAN Buses.
 - b) Explain the advantages of Internet enabled systems. How is the internet enabled device incorporated in embedded system?
8. Write short notes on any two of the followings:
 - i) JTAG
 - ii) Logic Analyzers
 - iii) Rate Monotonic scheduling
 - iv) Priority inversion
